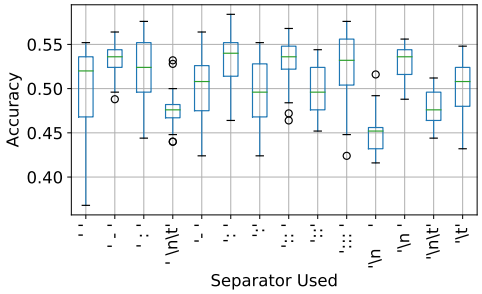

Figure 5: Example of accuracy variance for different choices of constants in $\mathcal{S}_1$ for task1283.

Table 1: Tasks where at least two constants yield different performance (*weakly* different if their boxes in a boxplot do not overlap, *strongly* if boxes including whiskers do not overlap).

	Median Spread (range [0, 1])	Weak Diff.	Strong Diff.
$\mathcal{C}$	0.144	29%	1%
$\mathcal{S}_1$	0.132	43%	22%
$\mathcal{S}_2$	0.238	0%	0%
$\mathcal{F}_{\text{item1}}$	0.176	0%	0%
$\mathcal{F}_{\text{item2}}$	0.173	45%	10%
$\mathcal{F}_{\text{casing}}$	0.188	3%	0%

Table 2: Examples of atomic changes' impact on accuracy using probability ranking (prefix matching shown in Table 4). {} represents a text field; p_2 yields higher accuracy than p_1 for all tasks.

Task Id	Prompt Format 1 (p_1)	Prompt Format 2 (p_2)	Acc p_1	Acc p_2	Diff.
task280	`passage:{}\n answer:{}`	`passage {}\n answer {}`	0.043	0.826	0.783
task317	`Passage::{} Answer::{}`	`Passage:: {} Answer:: {}`	0.076	0.638	0.562
task190	`Sentence[I]- {}Sentence[II]- {} -- Answer\t{}`	`Sentence[A]- {}Sentence[B]- {} -- Answer\t{}`	0.360	0.614	0.254
task904	`input:: {} \n output:: {}`	`input::{} \n output::{}`	0.418	0.616	0.198
task320	`target - {} \n{} \nanswer - {}`	`target - {}; \n{}; \nanswer - {}`	0.361	0.476	0.115
task322	`COMMENT: {} ANSWER: {}`	`comment: {} answer: {}`	0.614	0.714	0.100
task279	`Passage : {}. Answer : {}`	`PASSAGE : {}. ANSWER : {}`	0.372	0.441	0.069

11 non-multiple choice), sample 300 (p_1, p_2, p_3) triples for each, and the compute accuracy (using exact prefix matching) of each p_i on 250 samples. 32.4 and 33.6% of triples were monotonic for multiple-choice and non-multiple-choice tasks respectively. Given that random shuffling within a triple will result in monotonicity 33.3% of the time, this suggests that local search mechanisms like simulated annealing may not be effective as they require a locally smooth search space.

4.4 Prompt formats are identifiable transformations of prompt embeddings

Prompt format choices represent a deterministic transformation of the input, even if its impact on the resulting performance is hard to predict. We represent prompt embeddings as the last hidden layer obtained when processing the whole input prompt (immediately before generating the first token). We demonstrate that format choice yields a highly identifiable transformation over this embedding, which suggests that formats can be seen as transformations of the output probability distribution.

For each task, and for both $\{1, 5\}$-shot settings, we collect prompt embeddings from LLaMA-2-7B corresponding to 10 randomly sampled valid formats for 1000 evaluation examples. We train an XGBoost (Chen & Guestrin, 2016) classifier that maps from the top n principal components of a prompt embedding to the prompt format.[3] We find that although the original prompt embeddings are of size 4,096[4], using just the top 100 principal components can result in a classifier with $\geq$0.98 accuracy in format identification for all 31 tasks analyzed. Figure 7 shows the accuracy of format classification given a fixed number of principal components.[5] We find that classifier accuracy given just the top two components correlates moderately with the spread of performance in the prompts they represent (0.424, $p = 8.04 \cdot 10^{-6}$; 0.555 for the 5-shot setting; using exact prefix matching).

4.5 Fast exploration of the prompt formatting space: FormatSpread

In Section 4.2, we demonstrate that even when sampling just 10 formats from the space of plausible formats, we still observe significant performance spread on many tasks. However, this is only a lower

[3]We train with 800 vectors from each of the 10 formats (8000 vectors) and evaluate on the remaining 200.
[4]Equivalent to the dimension of hidden representations for LLaMA-2-7B.
[5]Figure 21 in the Appendix visualizes examples of the top two principal components for ten prompt formats.

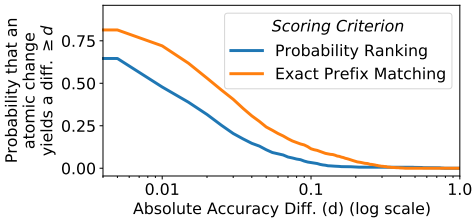

Figure 6: Probability that an atomic change (e.g. changing a space, separator) has a given impact in accuracy for two scoring criteria. 53 tasks, 30 sampled atomic changes each.

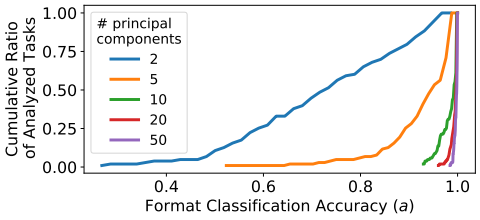

Figure 7: Cumulative ratio of tasks that can be classified with at most a accuracy using the top principal components of the last decoding layer of the prompt.

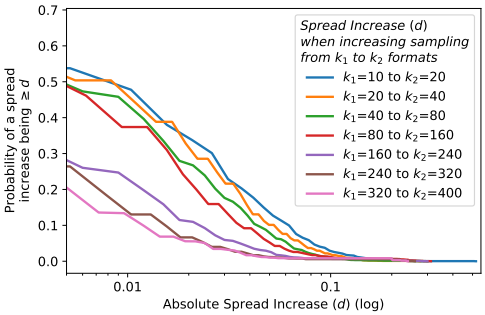

Figure 8: Probability of observing a spread increase of at least d when increasing sample size from k_1 to k_2 formats. 31 tasks, 100 trials each.

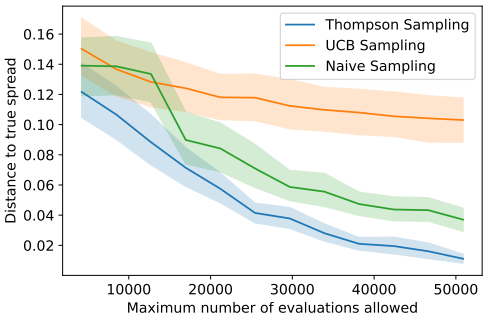

Figure 9: Difference between the true sample spread and each algorithm-found spread with respect to E (evaluation budget). 320 formats, $B = 20$, average of 5 trials over 31 tasks shown.

bound of the spread a task may exhibit when increasing the number of formats: for example, about 17% of tasks are expected to increase their spread by at least 5 accuracy points when increasing from 10 to 20 sampled formats. Figure 8 quantifies the expected increase in spread when increasing the number of formats by evaluating 500 formats on 250 samples each and computing expected gains.

Figure 9 compares the efficiency of Thompson sampling, UCB, and naive sampling for estimating spread with respect to a budget E (Section 3.2). To ensure accurate reports, we compute and show the true spread of the highest- and lowest-performing formats chosen by each method using all data. With a budget of 51,200 evaluations, Thompson sampling results in a spread within 1 accuracy point of the true spread, while naive sampling finds a spread within 4 points, and UCB within 11.

Finally, we use FORMATSPREAD to measure sensitivity of several models where inference is expensive. With a budget of 40,000 evaluations and 320 prompt formats, we find that 1-shot LLaMA-2-70B–ran using 4-bit quantization (Dettmers et al., 2022)–yields a median spread of 0.171 (mean=0.221, std=0.200, using probability ranking across 53 tasks; 25% of tasks had a spread of 0.292 or higher, with a maximum spread of 0.876), and GPT-3.5 yields a median spread of 0.064 (mean=0.110, std=0.115, across 53 tasks using exact prefix matching given that we do not have access to the full logits; 25% of tasks had a spread of 0.148 or higher, with a maximum spread of 0.562), showing sensitivity to formatting is still present even on larger models. 5-shot LLaMA-2-70B still shows high spreads, with 25% of tasks having a spread of 0.310 and a maximum of 0.841. See spread visualization in Figure 25, and a list of best and worst formats found in Table 6.

5 RELATED WORK

The task of automatically finding the best-performing prompt for a given task without changing model parameters has recently gained attention, given the constantly improving yet somewhat unpredictable performance of LLMs. Prior work has often focused on discovering optimal prompts with gradient-based methods, which are effective, but often lead to disfluent or unnatural prompts (Shin

et al., 2020), which can be mitigated with a Langevin dynamics-based method (Shi et al., 2022). Another approach is to learn, optimize, and insert continuous representations of prompts and tasks as input to models (Qin & Eisner, 2021; Lester et al., 2021; Ding et al., 2022; Ilharco et al., 2023). These methods also require access to the LLM's parameters, thus cannot be applied to models behind an API. In contrast, FORMATSPREAD does not assume access to any model internals. Prior gradient-free work has focused on edit-based enumeration over human-written prompts (Prasad et al., 2023), reinforcement learning (Deng et al., 2022), and by using LLMs themselves (Zhou et al., 2023; Gao et al., 2021). These works aim to achieve competitive task performance, even if the meaning of the prompt or instruction is modified. To our knowledge, we are the first to focus specifically on prompt formatting variance, a quintessential example of semantic equivalence.

Jailbreaking refers to the behavior of intentionally manipulating prompts to elicit inappropriate or sensitive responses, or otherwise reveal parts of the prompt that were intentionally not revealed. While the objective differs from our work, jailbreaking works (Wei et al., 2023; Zou et al., 2023) share the underlying technical question of finding the lowest-performing prompt. Our methods differ, since Wei et al. (2023) evaluate human-generated attacks to guide adversarial prompt design, and Zou et al. (2023) uses gradient-based search methods simultaneously across multiple models.

Some existing work has explored the influence of certain prompt design choices on model performance, for example the prompt's language (Gonen et al., 2022), the ordering of few-shot examples (Lu et al., 2022), and their patterns (Madaan et al., 2023). Other work has focused on providing textual interpretations of continuous prompt representations (Khashabi et al., 2022). Beyond autoregressive LLMs, existing work has focused on performance variance in masked language models (Elazar et al., 2021; Jiang et al., 2020). Our work follows efforts in other domains that explore the influence of spurious features on research evaluations, e.g., in deep reinforcement learning (Islam et al., 2017; Henderson et al., 2018) and statistical machine translation (Clark et al., 2011).

6 DISCUSSION

We introduce FORMATSPREAD, an algorithm that estimates the performance *spread* across prompt formatting choices.[6] We use FORMATSPREAD to evaluate the spread of several widely-used open-source LLMs for classification tasks in few-shot learning settings. We find that spread is large regardless of model choice, even when increasing model size, number of few-shots, or when using instruction tuning. FORMATSPREAD is designed to efficiently search the space of plausible prompt formats under a user-specified computational budget. For example, with a computational budget of exploring only 5% of the entire search space for task with 2,500 test examples and 320 plausible formats, we are able to estimate spread within 2 accuracy points of the true spread.

We also characterize the space of prompt formats, finding that it is largely non-monotonic and that few atomic features can be predictors of performance alone, although the separability of format embeddings is highly correlated with observed performance spread. These findings informed the design of our search procedure, where local search methods are not advantageous.

Our findings suggest that performance spread caused by arbitrary prompt formatting choices may influence conclusions made about model performance, especially when comparing models on benchmark tasks. Thus, we recommend that work evaluating LLMs with prompting-based methods would benefit from reporting a range of performance across plausible formats. However, we want to emphasize that single-format evaluation may still be sufficient for many use cases. For example, for researchers or practitioners who build systems on top of LLMs, choosing a single prompt format that works sufficiently well for use in this larger system is a valid methodological choice. However, we encourage future research to compute FORMATSPREAD when comparing their systems to out-of-the-box models, to ensure fair baseline representation. Furthermore, FORMATSPREAD can be used to identify lower-bound performance of a model or system. For example, when using a model for socially impactful tasks, such as stereotype classification in Figure 1, it is important to report the range of accuracy a non-adversarial user might encounter. Likewise, it is crucial to consider robustness to spurious features when claiming that models possess general abilities, such as theory of mind; and beneficial to report when e.g. exploring model biases. We leave it to future research to develop regularization procedures either during training or with an already-trained model to make models robust to diverse formatting choices.

[6]We thoroughly describe the limitations of our method in Appendix C.